

Enterprise Data Platform

Architecture Review Committee | Enterprise Data Management and Governance

Prepared by Placida Fernando
08/31/2026

Enterprise Data Platform Modernization

A strategic roadmap to a unified, governed data foundation supporting regulatory compliance, customer insight, and sustainable growth

This roadmap assesses the **current data landscape**, defines a **target-state architecture**, evaluates the **vendor and platform landscape**, and assesses **AI and automation readiness** - anchored in a trusted, governed enterprise data foundation.



Current State

Fragmented core, channel, and third-party sources; limited curation and inconsistent reporting.



Future State

Unified ingestion, curation, and consumption layers enabling governed, near-real-time analytics.



Vendor & Platform Landscape

Cloud-native warehouses, lakehouse platforms, hyperscalers, and turn-key solutions compared on integration, performance, and support.



AI & Automation Readiness

Maturity spectrum from AI-adjacent operational platforms to emerging innovation accelerators.



Trusted Enterprise Data Foundation

Integration

Batch & real-time

Performance

Scalability & security

Decisioning

Actionable insight

Support

Training & operations

Enterprise Data Platform to Support Regulatory, Customer, and Growth Outcomes

Vendors focus on Data Visualization and Actionable Insight

Turn-key/pre-built Solutions

- Trellance (rise Analytics), Snowflake, Lakehouse is on Snowpark, Azure – nightly batch, filters for new reports (not big on curate), limited reports, quick releases
- Lodstar - Mature CU EDW; strong board and portfolio reporting
- Whiteclay – Big on Profitability, Purpose-built CU DW; solid foundational schemas , lacking AI
- Klarivis - Very fast time-to-value; strong CU-specific models; intuitive, functions as an integration, analytics, and visualization platform
- Capitalize Analytics for implementation (possibly)

Enterprise Data & Cloud Platform Segmentation

1) Cloud-Native Data Platforms:

a) Data Warehouses with RPA -

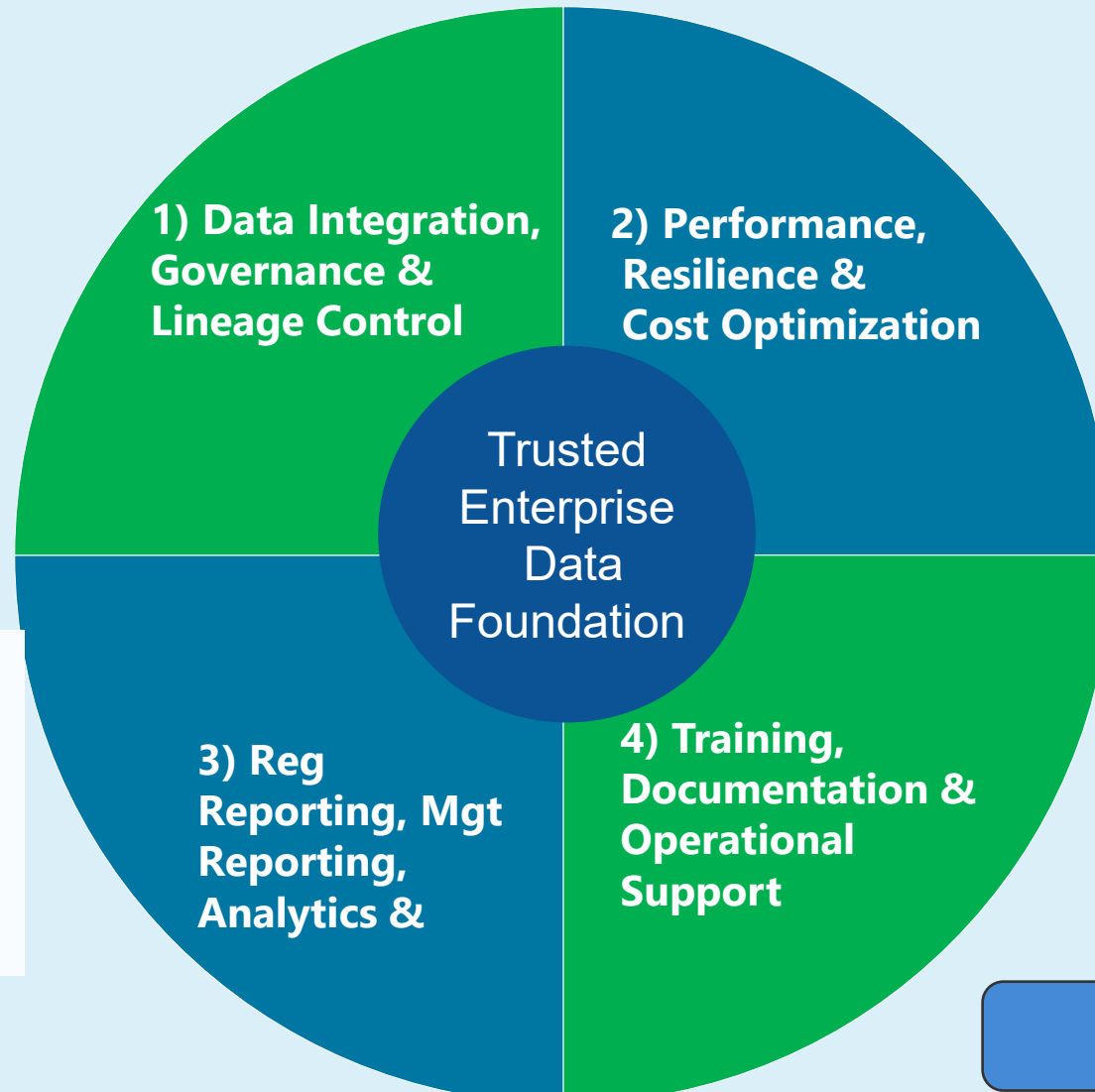
Snowflake, Google BigQuery, Amazon Redshift, Azure Synapse

b) Lakehouse/Analytics Platform – Databricks, MS Fabric suite

2) Hyperscale Cloud Providers: AWS, Azure, Google Cloud

3) Traditional / Hybrid Data Accelerators: SQL Server w/SSIS, Teradata, Oracle Heatwave (no need data replication-cheaper), IBM (relational – DB2, Hierarchical – IMS DB)

Designed to support regulatory compliance, customer insight, and sustainable growth.



Training, Operational Support

- Turn-key solution vendors will provide full support
- E-2-E vendors such as Capitalize Analytics

Actionable insights

- ML, NLP vs
- Generative and Agentic AI

Cost Optimization & Scalability

- Performance
- Security – RBAC, Data on private/public/hybrid cloud concerns
- Extensibility

Data Integration, Governance & Lineage Control

- Integrate Batch via ETL or real-time APIs
- Curate : Cleanse, Lineage
- Normalize into their standard/common data model
- Governance

Next-Generation Enterprise Data Platform Ecosystem : Outcome driven Open data accelerator

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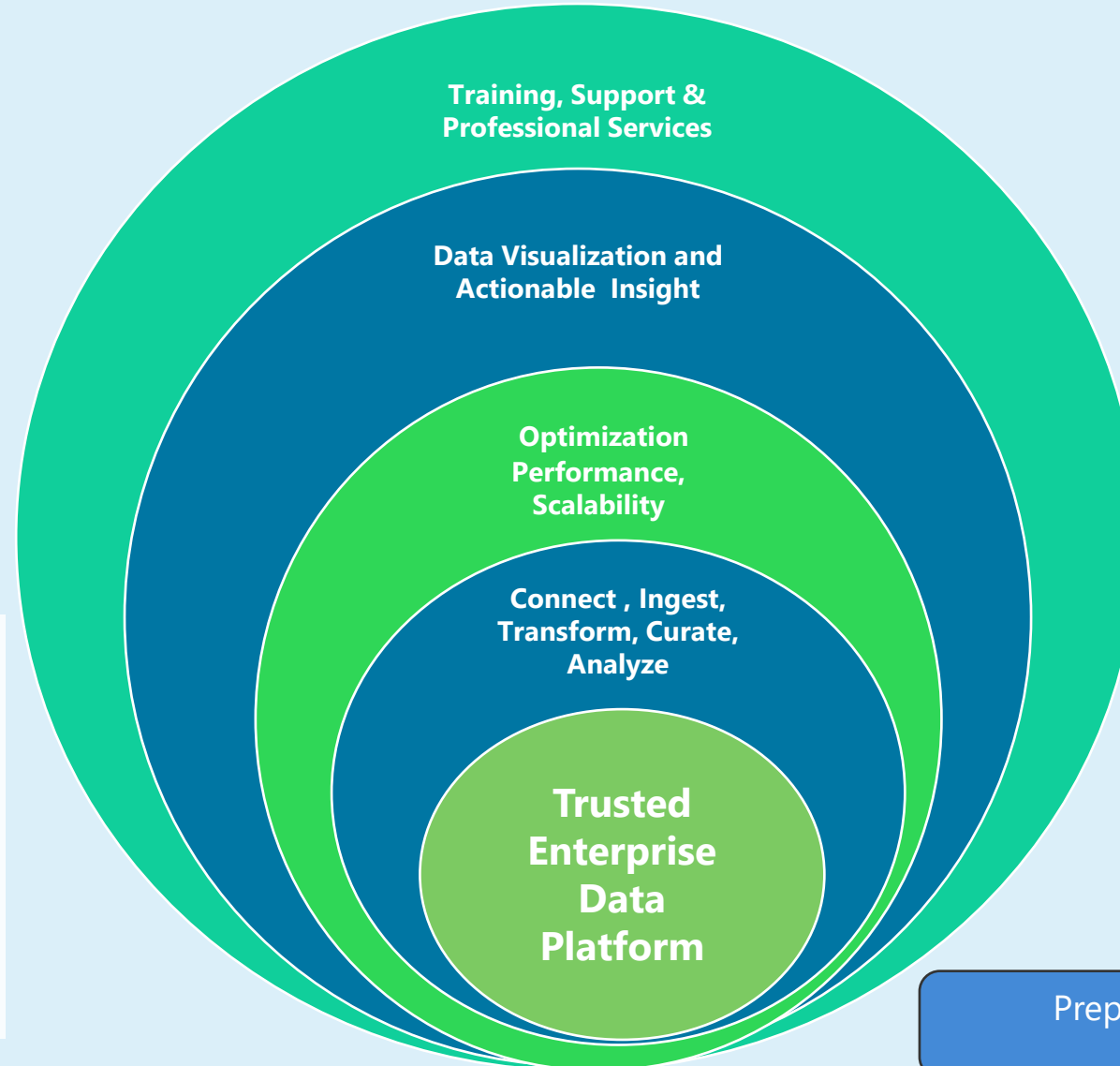
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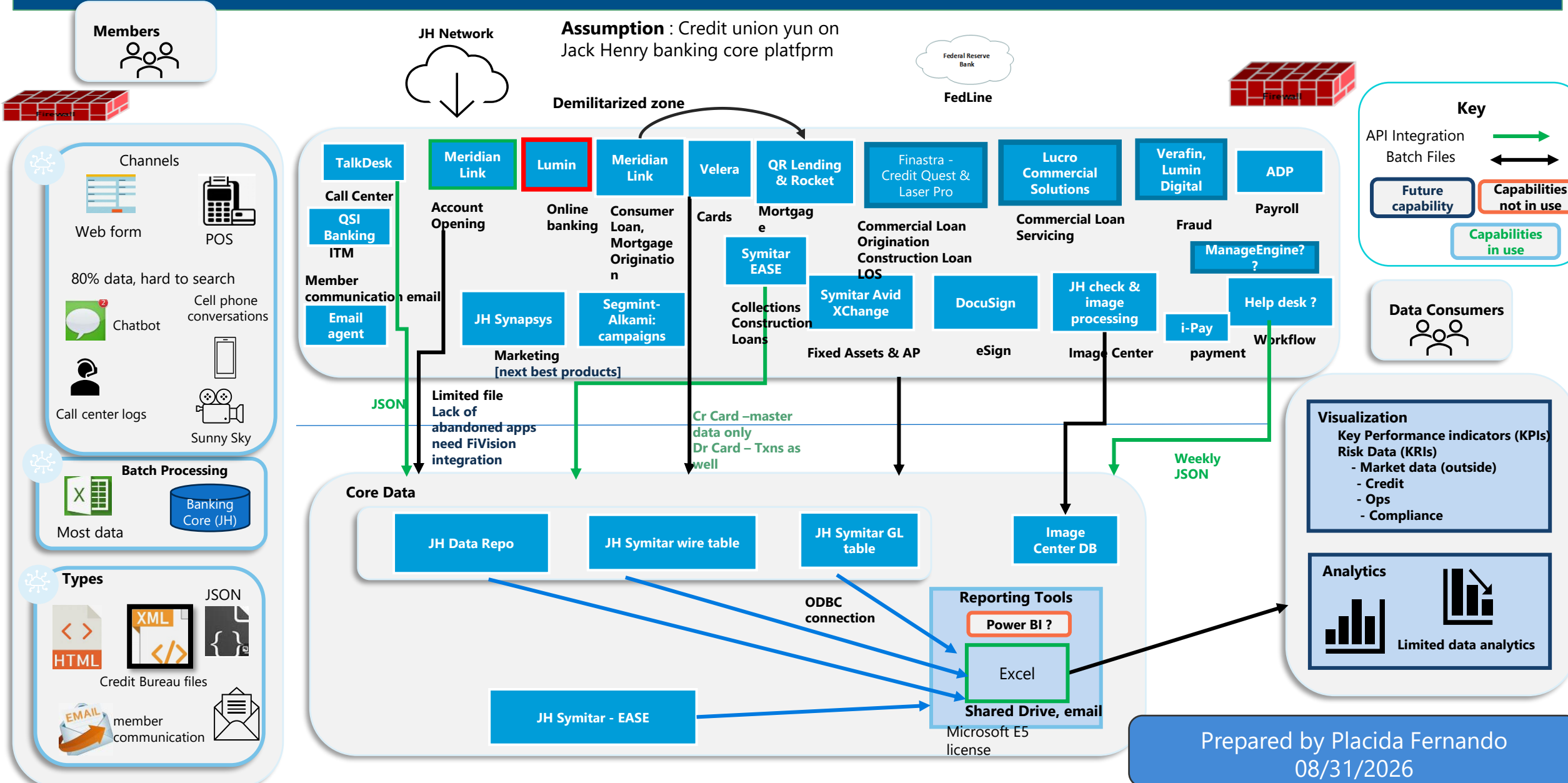
Optimization & Scalability

- Performance
- Security – RBAC, Data on private/public/hybrid cloud concerns
- Extensibility

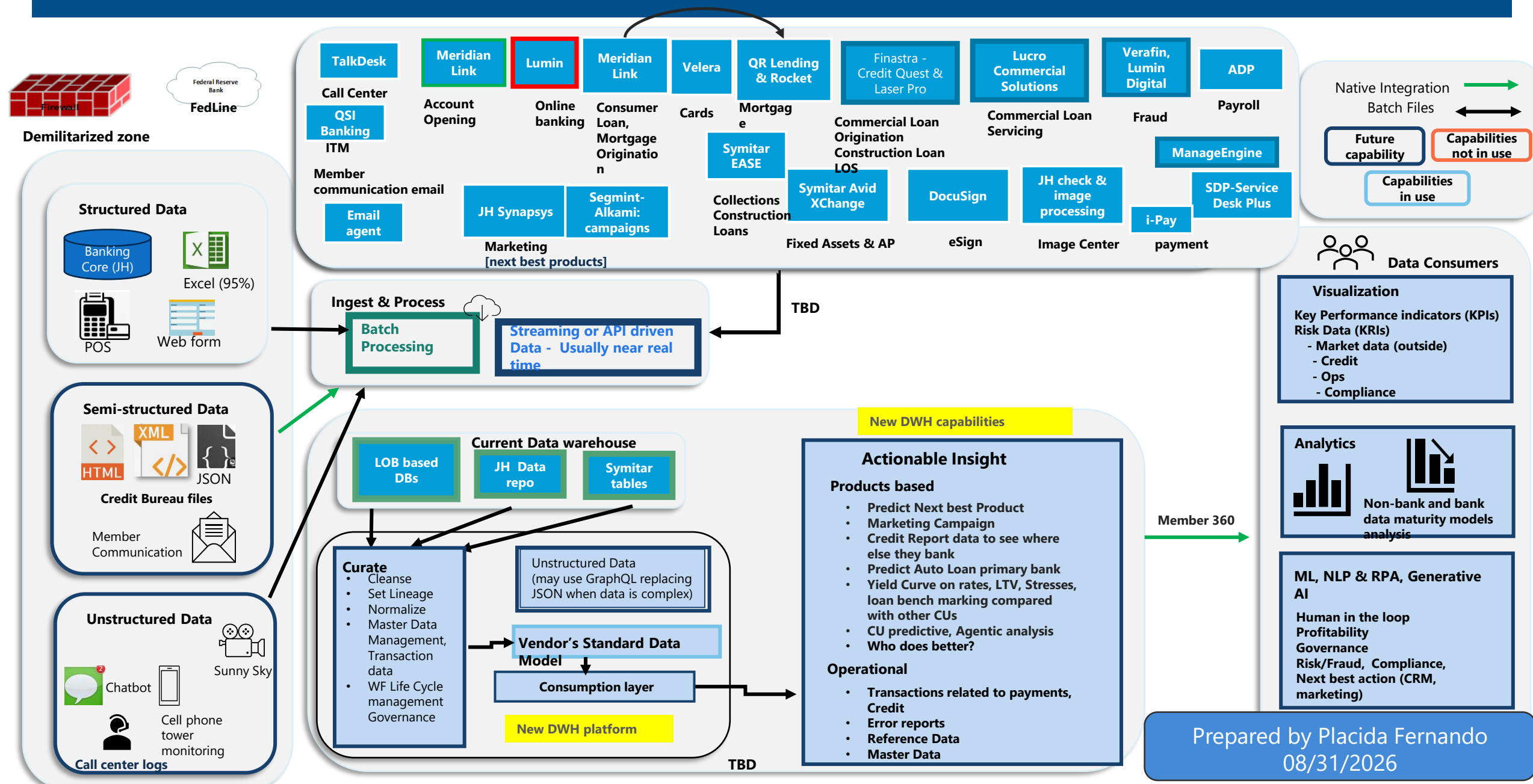
Connect, Ingest, Transform, Curate, Analyze

- Integrate Batch via ETL or real-time APIs
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Current State Architecture



Future State Data Platform Architecture



Additional Reference

AI for regional and CU Ecosystem

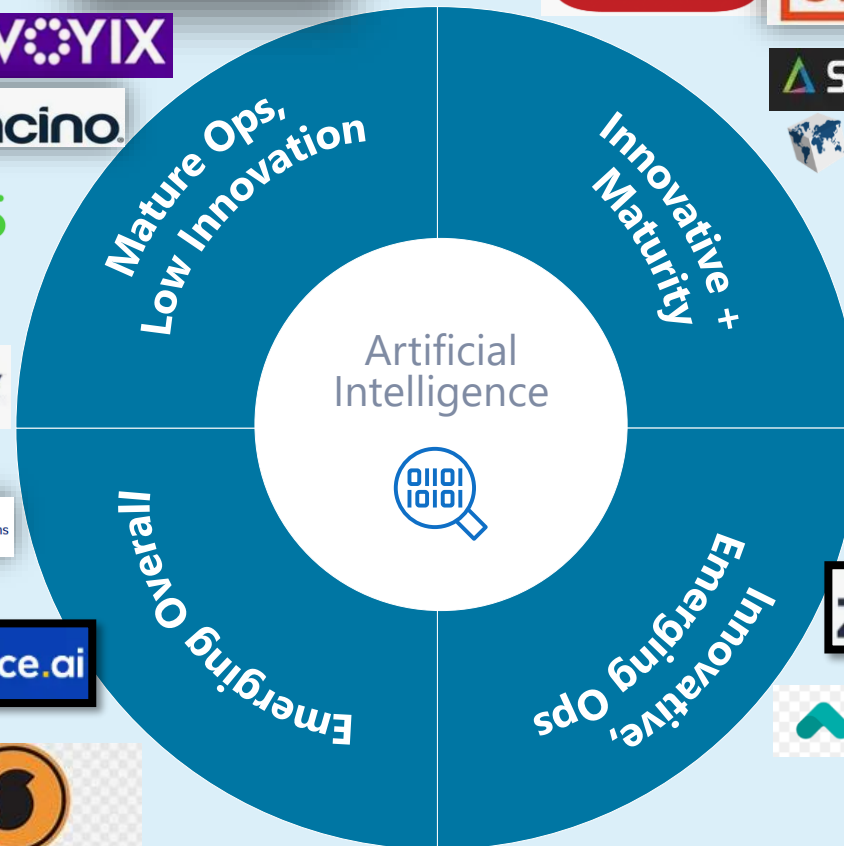
Loan automation,
analytics dashboards

- These platforms excel at **operational robustness, security, and regulatory alignment**, but they tend to be “AI-adjacent” rather than AI-native.
- They’re ideal for institutions prioritizing **risk management and reliability** over rapid innovation



Low-code tools, core
integration APIs.

These are **early-stage or niche AI/automation providers** that demonstrate conceptual innovation but lack proven scalability, regulatory readiness, or broad ecosystem support. Their technology may show promise in pilots or point use cases but hasn’t yet matured for mission-critical workloads.



Explainable AI, decision intelligence,
composable APIs.

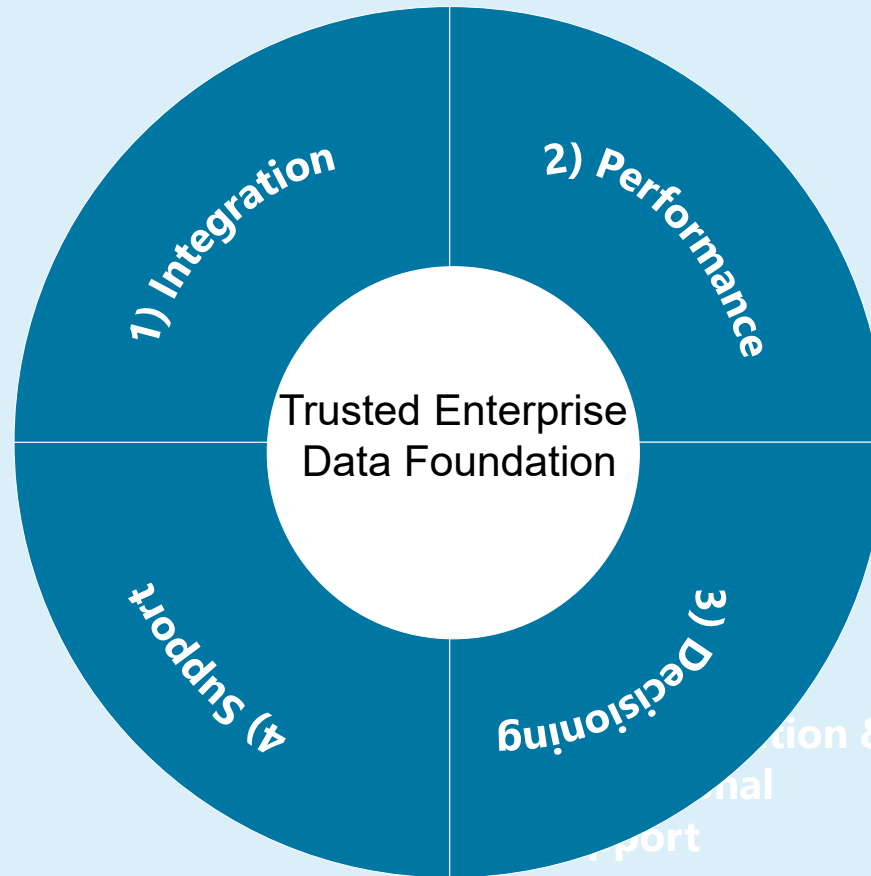
- “steady innovators” — platforms that regional banks can adopt with confidence to modernize today while building for tomorrow.
- They embody the architectural balance between stability, scalability, and adaptive intelligence.
- Mambu - Composable architecture, partner ecosystem, embedded finance enablement.

- represents the “innovation accelerators” — ideal for banks seeking differentiation through AI and automation, but requiring deliberate risk governance, pilot-first adoption, and strong integration controls before enterprise-wide scaling.

Generative AI pilots.

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